

# Priyal Parikh

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## EDUCATION:

**Master of Science**, Computer and Information Science  
**State University of New York at New Paltz, NY, USA.**  
**Bachelor of Engineering**, Computer Engineering  
**LDRP Institute of Technology and Research, GJ, India.**

August 2023 - May 2025

July 2019 - April 2023

## TECHNICAL SKILLS:

**Languages:** Python, JavaScript, TypeScript, C/C++, Java, SQL, Bash.

**Web & Full Stack Development:** HTML5, CSS3, React, Angular, Node.js, Django, FastAPI, Flask, Bootstrap 5, Tailwind CSS, REST APIs, Firebase, MongoDB, MySQL, PostgreSQL.

**AI/ML & Data Science:** TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face Transformers, OpenAI API, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook.

**DevOps & Cloud:** Docker, GitHub Actions (CI/CD), AWS (EC2, S3, Lambda), Microsoft Azure (basic), Linux/Unix.

**Tools & Platforms:** Git, GitHub, VS Code, Postman, ClickUp, Slack, Trello, Figma, Adobe XD, Jira, Microsoft Office Suite, Google Workspace, Agile/Scrum.

**Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Cybersecurity, Network Security, Advanced Data Structures, Web & Database Programming, Dynamic Programming, Discrete Structures.

## PROFESSIONAL EXPERIENCE:

**Full Stack, AI & ML Engineering Intern, SUPPLIFLEX (Remote, USA)**

July 2025 - Present

- Designed and deployed an AI-driven demand forecasting system using **Python, Flask, FastAPI, Prophet, MongoDB**, integrating predictions into **React** dashboards and achieving over **85%** accuracy while reducing stockouts.
- Built a Business Health Scorecard with **KPI tracking, real-time alerts, NLG summaries** using **Flask, React, Tailwind CSS**, enabling proactive decision-making with more than **70%** user engagement on AI-driven insights.
- Streamlined **CI/CD pipelines** by containerizing **Flask-based services** with **Docker**, deploying via **AWS Lambda & EC2**, and automating workflows using **GitHub Actions**, cutting release time by **50%** and maintaining **99% uptime**.

**Graduate Teaching Assistant, State University of New York at New Paltz, NY, USA**

December 2023 - May 2025

- Developed and launched an academic advising chatbot using **GPT-3.5-turbo, LangChain**, and **FastAPI**, hosted on **Firebase** with **Firestore**, decreasing advising workload by **70%**.
- Fine-tuned **GPT model** with custom academic datasets and developed domain-specific prompt strategies while integrating embedding-based retrieval using **Pinecone**, boosting content relevance and recommendation accuracy to **95%**.
- Led weekly lab sessions and offered personalized technical mentorship in **Python, Java, Data Structures and Algorithms**, mentoring **50+** students through debugging, project development, and exam preparation, significantly improving their academic performance.
- Engineered an automated class scheduling backend system for the Mechanical Engineering department using **Python, FastAPI**, and **Google Apps Script**, minimizing scheduling conflicts and manual effort.
- Designed a responsive frontend with **React** and **Tailwind CSS**, collaborating in an **Agile environment** and utilizing **GitHub** for version control.
- Implemented a student engagement dashboard leveraging the **Google Sheets API**, automating data extraction and visualization to analyze query trends and resolution times.

**Web Developer, INTECH, Gandhinagar, Gujarat, India**

July 2022 - June 2023

- Architected and deployed full-stack web applications using **Flask** and **Python**, streamlining internal workflows and enhancing system functionalities.
- Refactored **RESTful APIs** and optimized complex **SQL** queries, resulting in a **15%** reduction in backend response times.
- Strengthened backend capabilities in **Java** by implementing **multithreading, JDBC** for database access, and **servlet-based architectures**, applying **object-oriented design principles** to ensure code scalability and long-term maintainability.

## PROJECTS:

**Whisper Diary** | React | LLM | Chart.js

- Engineered a **React**-based mood journal integrating **AI-powered Sentiment Analysis** to interpret emotions from journal entries and generate personalized mental well-being recommendations.
- Created dynamic mood trend visualizations using **Chart.js** to help users monitor emotional patterns over time, boosting self-awareness and engagement.

**Chess180 AI** | React | Python | FastAPI | Reinforcement Learning | AI

- Designed and implemented an AI engine for Chess180, a custom chess variant with non-standard piece movements and win conditions, leveraging **Minimax** with **Alpha-Beta pruning** to optimize move selection and reduce computational overhead.
- Built a **Reinforcement Learning** system enabling the AI to improve through self-play and developed a **FastAPI** backend integrated with a **React** frontend, enabling seamless, real-time gameplay and user interaction.